

INTRODUCTION TO FATTY ACIDS (FA)

Definition:

- A fatty acid (FA) may be defined as an organic acid that occurs in a natural triglyceride.
- Monocarboxylic acid; chain length ranging from C4 to about 24 carbon atoms.
- FA are obtained from hydrolysis of fats.

Classification:

Saturated: No double bonds (Palmitic acid).

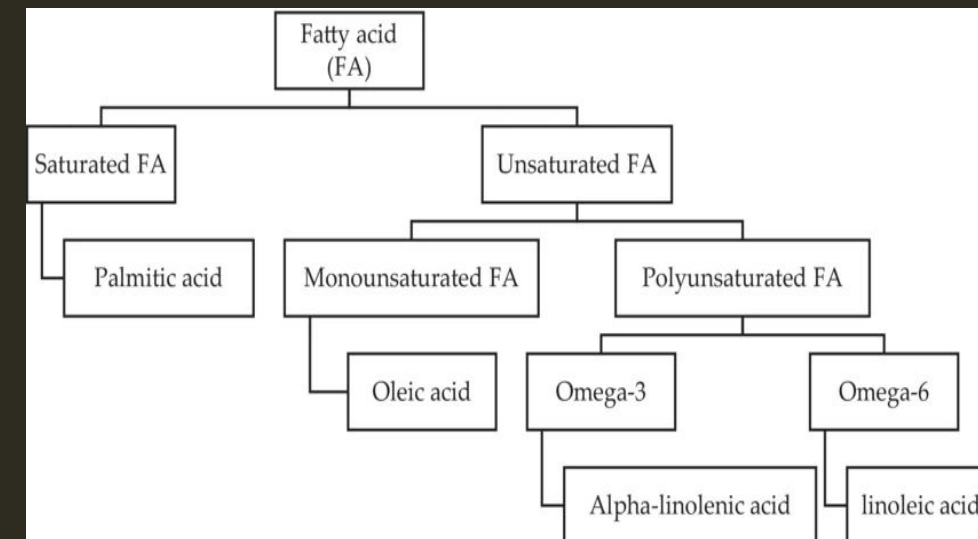
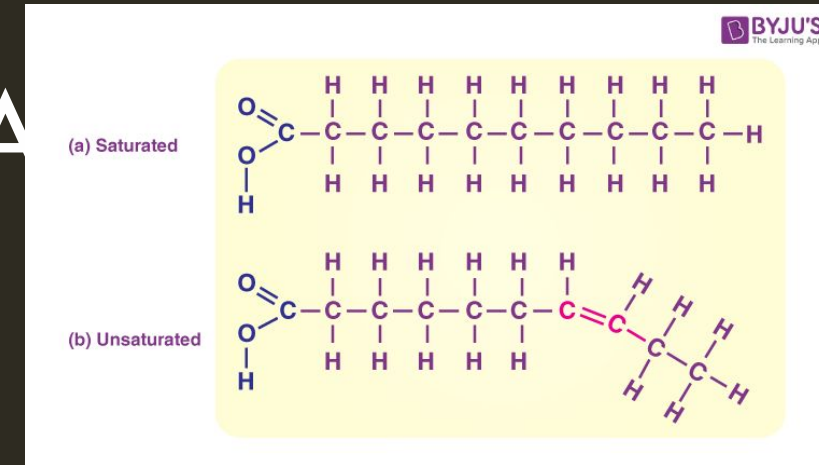
Unsaturated: contains double bonds (Oleic acid).

1. Monounsaturated fatty acids: only one double bond e.g. oleic acid.

2. Polyunsaturated fatty acids: more than one double bond. E.g. arachidonic acid, linoleic acid, etc

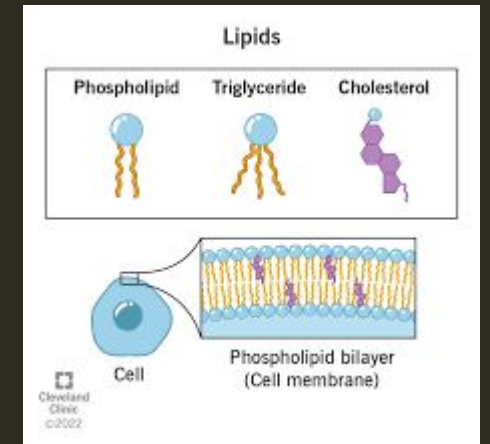
Note: Humans cannot synthesize ω -3 (alpha-linolenic) and

ω -6 (linoleic) fatty acids; they must be obtained from diet therefore, known as Essential FA.



PHYSIOLOGICAL ROLES OF FATTY ACIDS

- **Energy Reservoir**: FAs stored as Triacylglycerol (TAGs) in adipose tissue provide the highest energy yield (9.1 kcal/g).
- **Structural Components**: Major constituents of phospholipids and sphingolipids in cell membranes.
- **Metabolic Precursors**: FAs are the starting material for synthesizing ketone bodies and eicosanoids.
- **Protein Anchoring**: Covalent attachment of FAs (myristoylation/palmitoylation) anchors proteins to membranes.



CLINICAL SIGNIFICANCE: FATTY ACIDS IN DISEASE

- **Ketoacidosis:** Excessive β -oxidation (often in Type 1 Diabetes) leads to overproduction of ketone bodies.
- **Steatosis (Fatty Liver):** Imbalance between TAG synthesis and VLDL secretion.
- **Atherosclerosis:** High intake of trans-fatty acids and saturated fats correlates with elevated LDL (the "bad" cholesterol).

INTRODUCTION TO EICOSANOIDS

Definition: A family of signaling molecules (20-carbon compounds) derived primarily from **Arachidonic Acid**.

Nature: They act as "local hormones" (autocrine/paracrine signaling) with a very short half-life.

Main Classes:

Prostaglandins (PG)

Prostacyclins (PC)

Thromboxanes (TX)

Leukotrienes (LT)

THE SYNTHESIS PATHWAY (THE "CYCLOOXYGENASE" VS. "LIPOXYGENASE")

The Trigger:

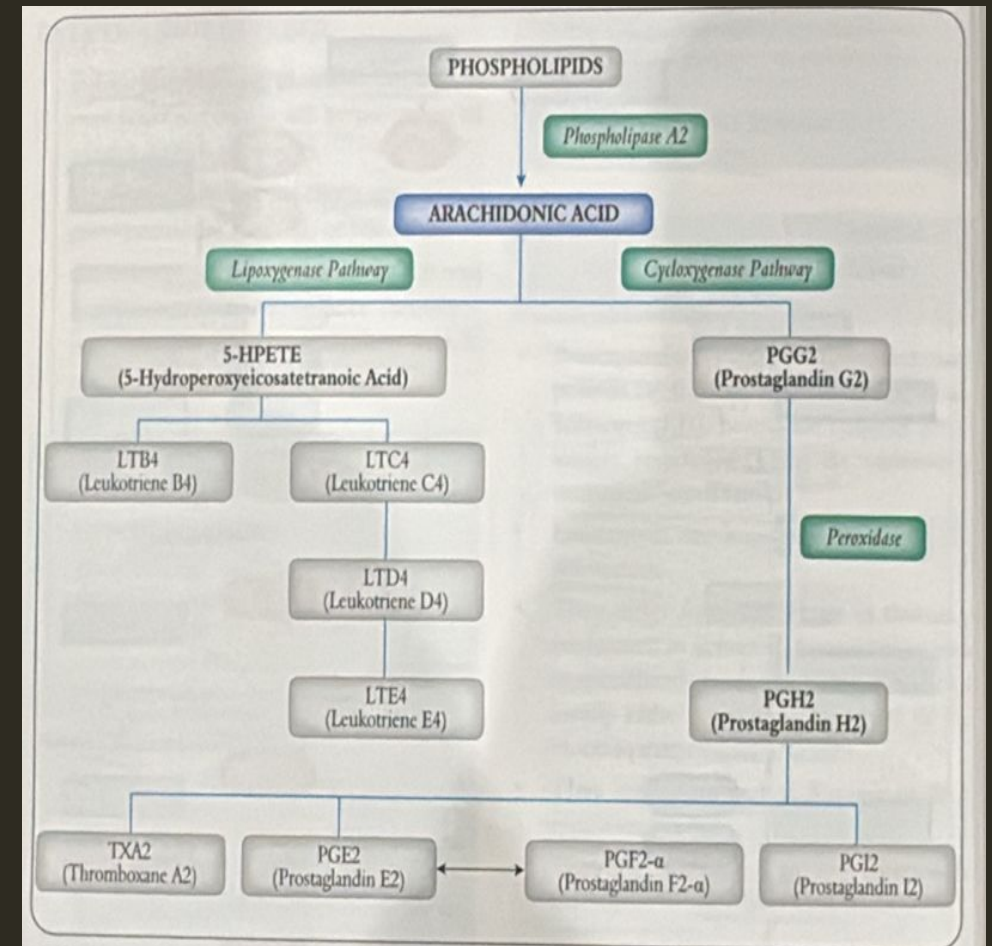
Phospholipase A₂ releases arachidonic acid from the membrane.

COX Pathway (Cyclooxygenase):

Leads to Prostaglandins (inflammation, fever, pain) and Thromboxanes (platelet aggregation).

LOX Pathway (Lipoxygenase):

Leads to Leukotrienes (allergic reactions, bronchoconstriction).



FUNCTION OF PROSTAGLANDINS

They have various role in:

- Inflammation
- Fever
- Regulation of blood pressure
- Blood clotting
- Immune system modulation
- Tissue growth
- Regulation of sleep/wake cycle

FUNCTION OF PROSTACYCLINS

- Synthesized by heart & vascular endothelial cells.
- Inhibit platelet and leukocyte aggregation
- induce vasodilation
- prevent clot formation
- decrease lymphocyte proliferation.

FUNCTION OF THROMBOXANES

- Synthesized by platelets
- Induce platelet aggregation
- promote vasoconstriction
- promote clot formation
- increase lymphocyte proliferation.

EICOSANOIDS IN HEALTH

- **Gastric Protection**: PGE₂ inhibits acid secretion and stimulates protective mucus.
- **Renal Function**: Maintains blood flow to the kidneys.
- **Vascular Tone**:
 - Prostacyclin (PGI₂)**: Vasodilation and inhibits platelet clumping.
 - Thromboxane (TXA₂)**: Vasoconstriction and promotes clumping.

Balance is key for cardiovascular health!

EICOSANOIDS IN DISEASE & PHARMACOLOGY

Asthma: Overproduction of Leukotrienes causes airway constriction.

Inflammation: Excess Prostaglandins mediate the redness, heat, and pain of injury.

Pharmacological Targets:

NSAIDs (Aspirin/Ibuprofen): Aspirin is an irreversible inhibitor. Aspirin targets the **Cyclooxygenase (COX-1 and COX-2)** enzymes. These enzymes are responsible for creating prostaglandins (which cause pain/inflammation) and thromboxane (which helps blood clot).

Steroids: Inhibit Phospholipase A2, blocking the entire cascade.

Features	COX-1(Constitutive)	COX-2(Inducible)
Expression	Constant; most tissues	Induced by inflammation/cytokines
Primary Role	Tissue homeostasis & protection	Mediating pain and inflammation
Gastric Effect	Protect gastric mucosa	Minimal role in stomach
Platelet Effect	Promotes aggregation(TX A ₂)	No effect on platelets
Inhibitors	Aspirin, naproxen, ibuprofen	Celecoxib
Clinical Risk	GI bleeding & ulceration	Potential cardiovascular risk

Drug Name	Selectivity	Binding Mechanism	Primary Clinical Use
Aspirin	Non-selective (COX-1 and COX-2)	Irreversible (Covalent)	Cardioprotection (anti-platelet), Pain
Ibuprofen	Non-selective	Reversible	General pain, Fever, Inflammation
Naproxen	Non-selective	Reversible	Long-acting pain relief (Arthritis)
Indomethacin	Non-selective	Reversible	Gout, Closure of Patent Ductus Arteriosus
Ketorolac	Non-selective (COX-1 preference)	Reversible	Short-term management of severe pain
Diclofenac	Non-selective (Slight COX-2 preference)	Reversible	Joint pain and inflammation
Celecoxib	Selective COX-2	Reversible	Osteoarthritis, Rheumatoid Arthritis
Etoricoxib	Highly Selective COX-2	Reversible	Chronic joint pain, Gouty arthritis
Meloxicam	Preferential COX-2	Reversible	Osteoarthritis (less GI side effects)
Etodolac	Preferential COX-2	Reversible	Post-operative pain, Arthritis

SUMMARY TABLE: LIPIDS AND HEALTH

Molecule	Health Role	Disease Link
Saturated FA	Membrane Structure	Heart Disease (High LDL)
omega-3 FA	Neuroprotection/Anti-inflammatory	Deficiency leads to poor growth
Prostaglandins	Cytoprotection (Stomach)	Inflammation & Chronic Pain
Leukotrienes	Immune response	Asthma & Anaphylaxis

LIPID STORAGE DISEASE



Gaucher's Disease



Tay-Sachs Disease

NIEMANN-PICK DISEASE

- **Enzyme deficiency:** sphingomyelinase (A/B)
- **Primary symptoms:**
 - enlarged organs,
 - difficulty swallowing,
 - progressive loss of motor skills.

GAUCHER DISEASE

- **Enzyme deficiency:** glucocerebrosidase
- **Primary symptoms:**
 - enlarged liver/spleen,
 - bone pain/fractures,
 - anemia,
 - low platelet counts.

Feature	Gaucher's Disease	Tay-Sachs Disease
Enzyme Deficiency	β -Glucocerebrosidase	Hexosaminidase A
Lipid Accumulation	Glucocerebroside	GM2 Ganglioside
Primary Systems	Liver, Spleen, and Bone	Central Nervous System (Brain)
Liver/Spleen Size	Enlarged (Hepatosplenomegaly)	Normal (No enlargement)
Neurological Loss	None in Type 1; Severe in Type 2	Rapid & Severe neurodegeneration
Eye Findings	None	'Cherry-red spot' on macula
Skeletal Impact	Bone pain, fractures, "Erlenmeyer flask" femur	None
Microscopic Clue	Gaucher Cells (Crinkled paper appearance)	Swollen neurons with "Onion-skin" lysosomes
Management	Enzyme Replacement Therapy (ERT)	Supportive/Palliative Care
Key Symptoms	Easy bruising, fatigue, bone pain, fractures	Progressive weakness, loss of motor skills, seizures, startle response

TAY-SACHS DISEASE

- **Enzyme deficiency**: hexosaminidase A
- **Primary symptoms**:
 - progressive neurological damage,
 - loss of motor skills,
 - seizures,
 - “cherry-red spot” in the eye.

CONCLUSION

- Fatty acids are not just "fat"; they are vital energy sources and precursors to potent signaling molecules.
- The balance of omega-3 vs. omega-6 and the regulation of eicosanoid pathways are central to modern pharmacology.

REFERENCES

- Lippincott's Illustrated Reviews: Biochemistry (8th Ed)
- Chatterjea's Textbook of Medical Biochemistry (8th Ed)

